

■ PromptMentor — Presentation & Pitch Guide

Hackathon Master Presentation Blueprint | Comprehensive Project Breakdown

■ Executive Summary (The 30-Second Elevator Pitch)

"PromptMentor is an interactive, real-time AI prompt coach and dual-LLM comparison workbench that teaches users how to write high-precision prompts instead of hiding fixes behind the scenes."

Most AI tools refine your prompt silently, leaving you dependent on black-box fixes. PromptMentor evaluates your draft across 6 core prompt engineering pillars, generates a live AI Quality Score (0–100), provides actionable feedback, and executes both original and refined prompts simultaneously so you visually see the quality difference.

■ Problem Statement vs. Our Solution

The Problem in Current AI Tools	PromptMentor's Innovative Solution
Black-box prompt auto-fixers modify prompts silently without teaching <i>why</i> .	Real-Time AI Coaching: Evaluates drafts across 6 prompt engineering pillars and provides live feedback.
Vague LLM responses caused by missing context, role, and explicit constraints.	Dual LLM Output Compare: Runs original vs. refined prompts in parallel with AI quality delta analysis.
Lack of measurable quality feedback for prompt writers.	AI Quality Score (0–100): Google Gemini (gemini-2.5-flash) dynamic radial quality meter.
No structured learning path for prompt engineering.	Interactive Flashcards & Leaderboard: 3D flip flashcards and community benchmark showcase.

■ Architecture & Core Component Breakdown

1. src/utils/geminiApi.js (AI Model Engine)

Uses Google Gemini API (gemini-2.5-flash) via structured JSON schema evaluation. Contains getEffectiveApiKey, coaxAnalyze, comparePromptOutputs, and a robust local heuristic fallback engine (generateFallbackCoaxAnalysis) that guarantees 100% uptime with 0 downtime.

2. src/components/PromptCoachingLeftPanel.jsx (Coach & Console)

Features Starter Preset Chips, a 6-Pillar Engineering Checklist (Clear Intent, Good Context, Specific Details, Has Constraints, Output Format, Role/Persona), Live Coach Advice card, and a ChatGPT-style prompt console input bar with a circular upward-arrow submit button.

3. src/components/ScoreRightPanel.jsx (Score Gauge & Dual Output Workbench)

Radial SVG gauge (0–100) colored dynamically (Emerald ≥85, Violet ≥70, Amber ≥50, Red <50), 5-pillar breakdown bars, Prompt Evolution Cards (Original vs Final), and Dual LLM Output Workbench rendering markdown responses with AI difference explanations.

4. src/pages/HomePage.jsx & Animated Background

Full-viewport breathing radial gradient powered by framer-motion and Canvas, featuring metallic multi-stop gradient headline typography and smooth staggered entrance animations.

5. src/components/Header.jsx & UI Components

Branded top navigation with penguin logo, PromptMentor title, API key status indicators, and ButtonColorful with custom ambient glow overlays.

■■ Step-by-Step Hackathon Presentation & Demo Script

Step 1: Hook the Judges (30 Seconds)

"Hello judges! Have you ever asked an AI model for help and received a generic, unhelpful answer? Current tools fix your prompt silently without teaching you anything. We built PromptMentor — a real-time collaborative AI prompt coach that grades your prompt, teaches key prompt engineering principles live, and demonstrates the exact quality gain side by side."

Step 2: Live Demo — Home Page (30 Seconds)

"On our Homepage, you'll notice our dark theme with a dynamic breathing radial gradient background. Users can jump straight into Prompt Improve, explore Interactive Flashcards, check the Leaderboard, or Track Progress."

Step 3: Core Demo — The Prompt Workbench (90 Seconds)

"In Prompt Improve, our console starts empty. Clicking 'Help me fix my code error' reveals that Context, Constraints, and Persona fail on our 6-Pillar Checklist. Clicking the upward arrow submit button calls Gemini, generating an AI Quality Score of 35/100 and live advice. Clicking 'Compare Live LLM Outputs' runs both prompts in parallel and renders side-by-side markdown outputs demonstrating zero guesswork."

Step 4: Technical Execution & Value (30 Seconds)

"PromptMentor is built with React 18, Vite, Tailwind CSS, Framer Motion, and Google Gemini API. If network issues or key limits occur, our local fallback engine ensures 100% demo uptime with zero downtime."

■ Anticipated Judge Questions & Bulletproof Answers

Q: How is this different from ChatGPT's built-in prompt enhancers or custom GPTs?

A: Custom GPTs auto-enhance prompts silently in a black box. PromptMentor is educational: it grades your prompt across 6 pillars, gives live advice, and displays dual LLM output comparison so you become skilled over time.

Q: What happens if the Gemini API key is missing or rate-limited?

A: PromptMentor includes a local heuristic fallback engine (generateFallbackCoaxAnalysis). It analyzes word count, intent, and keyword patterns locally to generate instant scores and suggestions with 0 downtime.

Q: How do you calculate the AI Quality Score (0–100)?

A: The score is judged dynamically by Gemini using a structured JSON prompt evaluation schema across 6 core pillars: Intent, Context, Specificity, Constraints, Format, and Role/Persona.